A FUZZY ART MAP NEURAL NETWORK SPEECH RECOGNITION SYSTEM BASED ON FORMANT RATIOS

by

Captain D.P. Young, CD, B.Sc., rrmc

Canadian Forces

A Thesis

Presented to the School of Graduate Studies

in the

Department of Electrical and Computer Engineering

The Royal Military College of Canada

Kingston, Ontario

In partial fulfillment of the requirements for the degree

Master of Engineering

May 1995

(c) Copyright by D.P. Young 1995

This thesis may be used within the Department of National Defense but copyright for open publication remains the property of the author.

ABSTRACT

This thesis investigates the applicability of a Fuzzy ART (Adaptive Resonance Theory) Map ANN (Artificial Neural Net) to the task of ASR (Automatic Speech Recognition). The motivation for this work stems from two observations. First, from an Internet discussion group survey, it is evident that the Fuzzy ART Map ANN is being applied to tasks in a variety of disciplines. These disciplines include Computer Science, Geology, and Medical Science, as well as Mechanical, Civil and Software Engineering. However, a survey of the literature has turned up no references to the application of the Fuzzy ART Map ANN to the ASR task. Secondly, ASR is a vital component of speech understanding systems that promise to improve the ease with which humans can interact with computers. Spoken communication with computers will allow users to remotely access computer based services through standard telephone links. This would make every telephone a potential computer terminal. As well, many of the world telephones use PC (Pulse Code) signaling and hence more sophisticated services that require DTMF (Dual Tone Multi-Frequency) signaling are not yet available. It is for these reasons that the development of ASR over telephone channels is imperative.

This thesis provides some basic background information about ANNs, ART, fuzzy logic, ASR, and formant ratio theory. It then describes how a Fuzzy ART Map ANN is implemented as the recognizer component of an ASR system to learn and identify formant ratio vectors extracted from spoken phrases. The audio frequency analysis is limited to a 300 to 3300 Hz bandwidth so that the system may operate on verbal commands over telephone channels. The recognition accuracy of this system is then tested using a database with eight phonetically rich phrases. The database was recorded by twelve males and two females over a four month period using a high quality microphone in a computer room environment. The tests are repeated using a multi-layer feedforward backpropagation ANN as the recognizer component to provide a relative performance benchmark. Finally, the performance of the Fuzzy ART Map ASR system is tested over the telephone channel.

ACKNOWLEDGMENTS

I wish to express sincere appreciation to my thesis advisor, Dr. Paul E. Allard, for his guidance and constructive criticism.

I would also like to thank the following people who volunteered their time and efforts to help build the RMC Speech Database that was used extensively throughout this thesis:

Ann Hopkinson, Bernard Tremblay, Colin Wortley, Danielle Bossart, David Gravel, Don Hamilton, Ian Hawes, Jean Dolbec, Liam Porter, Larry Reeves, Murray Johnson, Matthew Oliver, and Waldemar Borys.

Finally, I would like to acknowledge the support of my family -- Donna, Cassandra, Kody and Keegan. They have made this effort possible.

VITA

Name:

David Peter Young

Born:

Sudbury, Ontario, 1963

Son of Norman and Helen Young

Education:

Primary and Secondary School in Sudbury, Ontario. Completed Grade 13 in 1982. Royal Roads Military College, Victoria, British Columbia, 1982-86. Awarded Bachelor of Science Degree in Computer Science and Physics in 1986.

Experience:

Aerospace Engineering Officer, Canadian Forces from 1986 to date including:

- Programmer/Analyst, Aurora Software Development Unit (ASDU), CFB Greenwood, Nova Scotia, 1986.
- Staff Officer Automated Data for Aerospace Maintenance (ADAM), Air Command HQ, Winnipeg, Manitoba, 1987-88.
- Directing Staff Canadian Forces Officer Candidate School (CFOCS), CFB Chilliwack, British Columbia, 1988-90.
- Aircraft Maintenance Control and Records Officer (AMCRO), 419 Tac F(T) Sqn, CFB Cold Lake, Alberta, 1990-93.
- Post Graduate Student, Royal Military College of Canada, Kingston, Ontario, 1993-95

TABLE OF CONTENTS

Title Page i

Abstract ii

Acknowledgments iii

Vita iv

Table of Contents v

List of Figures x

List of Tables xii

One Introduction

1.1 Introductory Comments 1

1.2 Thesis Outline 3

Two Neural Networks Overview

2.1 Introduction 5

2.2 ANN Programs 7

2.3 The Artificial Neuron Model 7

2.4 Differentiating Between ANNs 10

2.4.1 ANN Architecture 10

2.4.2 ANN Learning Process 10

2.4.3 ANN Recall Process 11

2.4.4 Major ANN Architectures 13

2.5 Backpropagation Learning in Multi-layer Feedforward ANNs 14

2.6 Chapter Summary 15

Three The Fuzzy Art Map ANN

3.1 Introduction to Adaptive Resonance Theory 16
3.2 The Origin of ART Development 17
3.3 Competitive Learning Models 19
3.4 Stability Issues 21
3.5 The ART Cognitive Process 21
3.6 When Does Adaptive Resonance Occur? 23
3.7 The ART Advantage 24
3.8 Fuzzy Logic 25
3.8.1 Parallels in Fuzzy Logic and ANNs 25
3.8.2 The Proposition 26
3.8.3 The Possibility Distribution Defined By Proposition 27
3.8.4 The Quantified Proposition 27
3.8.5 Fuzzy Logic Applied To Adaptive Resonance Theory 28
3.9 The Fuzzy ART ANN 29
3.10 An Eight Stage Fuzzy ART Algorithm 30
3.11 The Fuzzy ART Map ANN 35
3.12 Chapter Summary 38

Four Automatic Speech Recognition

4.1 Introduction 39
4.2 ASR System Classification 39
4.2.1 Speaker Dependency 40
4.2.2 Manner of Speech 40
4.2.3 Active Vocabulary Size 41
4.3 A Four Stage ASR Process 42
4.4 Applicable ASR Research Areas 44

4.4.1 Rapid Training 44
4.4.2 Robust Speech Recognition 44
4.4.2.1 The Acoustic Environment 45
4.4.2.2 The Microphone System 45
4.4.2.3 The Transmission Channel 46
4.4.2.4 The Variability of Speech 47
4.5 Human Speech Communication 49
4.5.1 Speech Production 49
4.5.2 Speech Perception 50
4.6 Formant Ratio Theory 51
4.7 Chapter Summary 55

Five A Fuzzy Art Map ASR System

5.1 Introduction 57
5.2 Stage #1 -- Speech Recording 58
5.3 Recording Method and Apparatus 60
5.4 Storage Method 62
5.5 Stage #2 -- Signal Processing 67
5.6 Stage #3 -- Feature Extraction 70
5.6.1 Noise Masking 70
5.6.2 Silence Reduction 71
5.6.3 Frequency Amplitude Normalization (Pre-emphasis) 71
5.6.4 Peak Picking 73
5.6.5 Formant Screening 73
5.7 Feature Representation 75
5.7.1 Tri Formant Vector (TFV) Representation 75
5.7.2 Bi Formant Ratio Vector (BFRV) Representation 77
5.7.3 Normalized Polar Angle (NPA) Representation 77
5.7.4 Tri Formant Ratio Vector (TFRV) Representation 79

5.8 Stage #4 -- Utterance Classification 79
5.8.1 Modifications to the Fuzzy ART Map ANN 83
5.8.2 Network Training 83
5.8.3 Information Retrieval 85
5.9 Chapter Summary 86

Six Experimental Results

6.1 Introduction 87
6.2 Testing Goals 87
6.2.1 Testing Control Measures 88
6.2.2 Testing Methods 88
6.3 Test Results 91
6.4 Evaluation Of Results 97
6.4.1 Benchmark ANN Performance Comparison 98
6.4.2 Speaker Dependency Performance Comparison 98
6.4.3 Transmission Channel Performance Comparison 98
6.4.4 Feature Representation Performance Comparison 99
6.4.5 Error Analysis 99
6.5 Chapter Summary 102

Seven Concluding Remarks

7.1 Thesis Summary 103
7.2 Conclusions 104
7.3 Future Work 105

References 106

Annexes

Annex A ASR System Hardware and Software

Annex B ASR Demo Program Code (*Stdemo Player 1.5*)

Annex C MAKESPEC.M Program Code (*Matlab 4.0*)

Annex D FORMANT3.M Program Code (*Matlab 4.0*)

Annex E FARTMAPT.M Program Code (*Matlab 4.0*)

Annex F BPMAPPER.M Program Code (*Matlab 4.0*)

Annex G BPTESTER.M Program Code (*Matlab 4.0*)

LIST OF FIGURES

Figure 2.1	Simplified Biological Neural Network	5
Figure 2.2	A Basic Artificial Neuron Model	9
Figure 2.3	Typical Neuron Activation Functions	9
Figure 2.4	ANN Architecture Nomenclature	12
Figure 3.1.A	Serial List Learning Error -- Experimental Results	18
Figure 3.1.B	Serial List Learning Error -- Classical Theory Prediction	18
Figure 3.2	Competitive Learning Model	20
Figure 3.3	The Fuzzy ART ANN Model	34
Figure 3.4	Fuzzy ART Map Architecture	37
Figure 4.1	A Four Stage ASR Process	43
Figure 4.2	A Speech Variability Pyramid	48
Figure 4.3	Vowel Formant Frequencies from Table 4.2	54
Figure 4.4	Vowel Formant Ratios from Table 4.2	54
Figure 4.5	Normalized Polar Angle Formant Ratios of Table 4.2 Vowels	55
Figure 5.1	RMC Speech Database File Naming Convention	63
Figure 5.2	ASR System Hardware Components	65
Figure 5.3-M	Microphone Speech Signal MDY12204.WAV	66
Figure 5.3-T	Telephone Speech Signal TDY12204.WAV	66
Figure 5.4-M	Chunk #1 of MDY12204.WAV	68
Figure 5.4-T	Chunk #1 of TDY12204.WAV	68
Figure 5.5-M	Chunk #2 of MDY12204.WAV	68
Figure 5.5-T	Chunk #2 of TDY12204.WAV	68
Figure 5.6-M	Chunk #3 of MDY12204.WAV	68
Figure 5.6-T	Chunk #3 of TDY12204.WAV	68
Figure 5.7-M	Spectrogram of MDY12204.WAV	69
Figure 5.7-T	Spectrogram of TDY12204.WAV	69
Figure 5.8-M	Microphone Noise Mask	69
Figure 5.8-T	Telephone Noise Mask	69
Figure 5.9-M	Noise Masking of MDY12204.WAV	69

Figure 5.9-T	Noise Masking of TDY12204.WAV	69
Figure 5.10-M	STE Curve for MDY12204.WAV	72
Figure 5.10-T	STE Curve for TDY12204.WAV	72
Figure 5.11-M	Silence Reduced MDY12204.WAV	72
Figure 5.11-T	Silence Reduced TDY12204.WAV	72
Figure 5.12-M	Octave Enhanced MDY12204.WAV	72
Figure 5.12-T	Octave Enhanced TDY12204.WAV	72
Figure 5-13-M	Local Maxima in MDY12204.WAV	74
Figure 5-13-T	Local Maxima in TDY12204.WAV	74
Figure 5.14-M	Formants in MDY12204.WAV	74
Figure 5.14-T	Formants in TDY12204.WAV	74
Figure 5.15-M	Formant Vectors in MDY12204.WAV	74
Figure 5.15-T	Formant Vectors in TDY12204.WAV	74
Figure 5.16-M	Tri Formant Vector Representation of MDY12204.WAV	76
Figure 5.16-T	Tri Formant Vector Representation of TDY12204.WAV	76
Figure 5.17-M	Bi Formant Ratio Vector Representation of MDY12204.WAV	78
Figure 5.17-T	Bi Formant Ratio Vector Representation of TDY12204.WAV	78
Figure 5.18-M	Tri Formant Ratio Vector Representation of MDY12204.WAV	80
Figure 5.18-T	Tri Formant Ratio Vector Representation of TDY12204.WAV	80
Figure 5.19-M	Fuzzy ART Map Recognition of MDY12204.WAV	81
Figure 5.19-T	Fuzzy ART Map Recognition of TDY12204.WAV	81
Figure 5.20	ASR System Software Components	82
Figure 5.21	Modified Fuzzy ART Map ANN	84
Figure 6.1	Benchmark ANN -- Sample Training Progress Record	101
Figure 6.2	Benchmark ANN -- Sample Training Error Summary	101

LIST OF TABLES

Table 2.1	Comparison of Conventional and Neural Network Computation	8
Table 2.2	Some Major ANNs Classified By Learning and Recall Modes	13
Table 3.1	ART Architecture Learning Properties	24
Table 3.2	Fuzzy Logic Compared To Binary Logic	25
Table 3.3	An Eight Stage Fuzzy ART Algorithm	33
Table 4.1	Performance Classification Of Some Commercial ASR Systems	41
Table 4.2	A Comparison of Vowels by Formant Frequencies and Formant Ratios.....	53
Table 5.1	RMC Speech Database Speaker Profiles	59
Table 5.2	RMC Speech Database Phrase List	60
Table 6.1	Benchmark ANN / BFRV / Microphone / Speaker Dependent	92
Table 6.2	Benchmark ANN / BFRV / Microphone / Speaker Independent	92
Table 6.3-M	Fuzzy ART Map / BFRV / Speaker Dependent / Microphone	93
Table 6.3-T	Fuzzy ART Map / BFRV / Speaker Dependent / Telephone	93
Table 6.4-M	Fuzzy ART Map / BFRV / Speaker Independent / Microphone	94
Table 6.4-T	Fuzzy ART Map / BFRV / Speaker Independent / Telephone	94
Table 6.5-M	Fuzzy ART Map / TFRV / Speaker Dependent / Microphone	95
Table 6.5-T	Fuzzy ART Map / TFRV / Speaker Dependent / Telephone	95
Table 6.6-M	Fuzzy ART Map / TFRV / Speaker Independent / Microphone	96
Table 6.6-T	Fuzzy ART Map / TFRV / Speaker Independent / Telephone	96
Table 6.7	ASR System Testing Summary	97